



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Review the scenario below. Then complete the step-by-step instructions.

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets

- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below. You can explore them here:

- **Identify** security risks through regular audits of internal networks, systems, devices, and access privileges to identify potential gaps in security.
- **Protect** internal assets through the implementation of policies, procedures, training and tools that help mitigate cybersecurity threats.
- **Detect** potential security incidents and improve monitoring capabilities to increase the speed and efficiency of detections.
- **Respond** to contain, neutralize, and analyze security incidents; implement improvements to the security process.

Recover affected systems to normal operation and restore systems data and/or assets that have been affected by an incident.

Summary	The company faced a security issue when all network services stopped working. The team discovered it was a DDoS attack using many ICMP packets. They responded by blocking the attack and turning off non-critical services so that the important services could be restored.
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Identify	Attackers targeted the company with an ICMP flood. The whole network was affected, and all critical systems had to be secured and restored.
Protect	The cybersecurity team added a firewall rule to limit incoming ICMP packets and used an IDS/IPS to block suspicious ICMP traffic.
Detect	The cybersecurity team set the firewall to check for fake IP addresses in incoming ICMP packets and used network monitoring to spot unusual traffic.
Respond	In future security events, the team will isolate affected systems to stop further disruption, restore critical services, review network logs for suspicious activity, and report incidents to management and, if needed, legal authorities.
Recover	To recover from an ICMP flood DDoS attack, network services should be restored step by step. First, block the attack at the firewall. Stop non-critical services to reduce traffic, then restore critical services. Once the attack ends, non-critical systems can be turned back on.

Reflections/Notes: The team restored critical services quickly and blocked the attack. Next time, earlier detection and automated DDoS blocks could help. Regular training and testing will improve response. This shows the value of prepared plans and monitoring.